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SUPERSEDING
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FEDERAL SPECIFICATION

CLOTH, BIRDSEYE AND GAUZE: COTTON

This specification was approved by the Assistant Administrator, Office of Federal Supply and Services, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for two types of cotton cloth.

1.2 Classification.

1.2.1 Types and classes. The cloth shall be of the following types and classes as specified (see 6.2):

Type I	-	Birdseye
Class 1	-	Diaper cloth (bleached)
Class 2	-	Coverall cloth (bleached)
Class 3	-	Coverall cloth (dyed)
Type II	-	Gauze - double layer (bleached)

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specification:

PPP-P-1134	-	Packaging and Packing of Cotton and Cotton-Synthetic Fiber Blend Fabrics (Excluding Duck Fabrics)

Federal Standards:

FED-STD-4	-	Glossary of Fabric Imperfections
FED-STD-191	-	Textile Test Methods

FSC 8305

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Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of Military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Laws and Regulations:

Rules and Regulations Under the Textile Fiber Products Identification Act

(Copies may be obtained without charge from the Federal Trade Commission, Washington, DC 20580.)

3. REQUIREMENTS

3.1 Standard sample. When a standard sample is available, the finished cloth shall match the standard sample for shade and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.2 Material (see 6.6).

3.2.1 Cotton. The yarn shall be made from cotton and shall be carded, drawn, and spun.

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3.3 Color.

3.3.1 Bleached. The color for type I, classes 1 and 2 and type II cloth shall be chemically bleached to match the available standard sample (see 6.4). An optical brightener which flouresces to the blue region may be used unless otherwise specified. The bleached finished cloth shall not discolor to a degree greater than that shown by the standard sample when tested as specified in 4.2.3. When no standard sample is available, the discoloration shall be not less than the rating "fair" when tested as specified in 4.2.3.

3.3.2 Dyed. The color for type I, class 3 cloth shall be Olive Green 107 or as specified (see 6.2 and 6.4). The use of dyes or substances containing elementary sulfur compounds capable of oxidation to sulfuric acid is prohibited. The dyed and finished cloth shall contain no more labile sulfur than that shown by the standard sample when tested as specified in 4.2.3. When a standard sample is not available, the dyed and finished cloth shall show not more than a slight trace of labile sulfur as defined in the test method when tested as specified in 4.2.3.

3.3.3 Matching. The color shall match the available standard sample under artificial daylight having a color temperature of 7000 ± 500 K and shall have a good approximation to the standard sample under incandescent lamplight at 2850 ± 100 K.

3.3.4 Colorfastness. The dyed cloth shall show colorfastness to laundering (after 3 cycles), perspiration, crocking, and light equal to or better than the standard sample. When no standard sample is established or when the standard sample is not a reference for colorfastness, the dyed cloth shall show "good" colorfastness to laundering (after 3 cycles), perspiration, and light, and no more crocking than Munsell Value 8.5 when tested as specified in 4.2.3.

3.4 Physical requirements. The finished cloth shall conform to the requirements specified in table I or table II as applicable when tested as specified in 4.2.3.

TABLE I. Physical requirements - Type I, birdseye

Class	Weight per square yard (ounces)		Width inches (including selvage)		Yarns per inch (minimum)		Breaking strength (pounds) minimum	
	Min.	Max.	Min.	Max.	Warp	Filling	Warp	Filling
1	4.0	---	26-1/2	27-1/2	60	43	40	50
2	3.4	3.8	as specified		60	46	40	50
3	3.7	4.2	as specified		60	46	40	50

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TABLE II. Physical requirements - Type II, gauze

Weight 1/ (oz/sq. yd.)	Width (including selvages) inches		Yarns per inch (sum of warp and filling) 2/	Breaking strength 1/ (pounds) minimum	
Minimum	Min.	Max.	Minimum	Warp	Filling
3.2	39-1/2	40-1/2	74	45	25

1/ Utilizing two plys of panel.

2/ Determined on single ply of panel, but both single panels shall conform to this requirement.

3.4.1 Weave.

3.4.1.1 Type I, classes 1, 2, and 3. The cloth shall be a single layer cloth in a birdseye weave.

3.4.1.2 Type II. Type II cloth shall be a double layer gauze cloth woven in a plain interlocking-type weave. The combining, or interlocking, of the double layer to form a single layer shall be throughout the length of the cloth and at regular intervals filling-wise to form pinking bars.

3.4.1.2.1 Pinking bar. A filling-wise bar (pinking bar) 1-1/2 ± 1/2 inches wide shall be woven into the cloth at regular intervals so as to form panels of the double layer (2-ply cloth). The 1-1/2 inch pinking bars shall be made by weaving the warps of the 2-ply cloth to form a single layer cloth.

3.4.1.2.2 Panels. The panels of 2-ply cloth formed by the pinking bars shall measure 19 ± 1/2 inch in length from inside of pinking bar to inside of pinking bar.

3.4.1.2.3 Folding stripes. The supplier may, at his option, insert folding stripes. If folding stripes are used, one stripe consisting of four ends weaving as one in a group of 3 or 4 shall be positioned 13-1/2 ± 1/2 inch from one selvage. Another stripe shall be positioned 8 ± 1/2 inch from the opposite selvage. The folding stripes shall become part of the single layer pinking bars at point of intersection.

3.5 Shrinkage. The sum of the warp and filling residual shrinkage for type I, class 1, or type II cloth shall not exceed 20 percent when tested as specified in 4.2.3. The residual shrinkage for type I, classes 2 and 3 cloth shall not exceed 2.0 percent in either the warp or filling directions when tested as specified in 4.2.3.

3.6 Absorbency. Type I, class 1 and type II cloth shall show a minimum absorption of 60 percent when tested as specified in 4.2.3. Type I, classes 2 and 3 cloth shall show a minimum absorption of 65 percent when tested as specified in 4.2.3.

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3.7 Finish. No additive finishes shall be applied and the character of the finished cloth shall be equal to that of the standard sample.

3.7.1 Nonfibrous materials. The starch and protein content including chloroform-soluble and water-soluble material of the finished cloth shall not exceed 2.0 percent when tested as specified in 4.2.3 except that the determination for type I, class 3 shall be prior to dyeing.

3.8 pH. The pH value of the water extract of the finished cloth shall be not less than 5.5 nor more than 9.5 when tested as specified in 4.2.3.

3.9 Length and put-up. Unless otherwise specified (see 6.2), the cloth shall be furnished in continuous lengths, each not less than 30 yards (see 6.5). The cloth shall be put up on full width rolls as specified in PPP-P-1134.

3.9.1 Pieces. Pieces of type II cloth shall begin and end with a pinked edge located in the middle of the pinking bar.

3.10 Fiber identification. Each roll shall be labeled or ticketed for fiber content in accordance with the Rules and Regulations Under the Textile Fiber Products Identification Act.

3.11 Workmanship. The finished cloth shall conform to the quality of product established by this specification. The occurrence of defects shall not exceed the levels set by the applicable acceptance quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

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4.2.2 End item examination. The end item shall be examined in accordance with 4.2.2.1 through 4.2.2.3.2.

4.2.2.1 Yard-by-yard examination. The required yardage of each roll shall be examined for the defects listed below. All defects which are clearly noticeable at normal inspection distance (3 feet) shall be counted regardless of their proximity to each other except that when two or more defects represent a single local condition of the cloth, only one shall be scored. A continuous defect shall be counted as one defect for each yard or fraction thereof in which it occurs. The sample unit shall be 1 yard, full width of the finished cloth. The lot size unit shall be 1 yard. The inspection level shall be I and the acceptable quality level, expressed in terms of defects per hundred units, shall be 6.5. The number of rolls from which the sample is to be selected shall be in accordance with table III. An approximately equal number of yards shall be examined from each roll in the sample.

Defects
Hole, cut, or tear
Smash
Knot 1/
Grease or oil stain
Embedded crease or wrinkle

1/ To be scored only when larger than limit shown for knots on figure 1 of FED-STD-4.

4.2.2.2 Overall examination. Each roll selected for yard-by-yard examination in accordance with table III shall be examined over its entire length for the defects listed below. Each defect shall be counted not more than once in each roll examined. If the total number of defects in the sample rolls exceeds the maximum number specified in table III, the lot shall be rejected.

Defects
Selvage curled, folded, rolled, or slack continuously or intermittenly throughout the roll
Over-all uncleanliness 1/
Objectionable odor
Uneven weaving throughout the roll
Not bleached white (type I, classes 1 and 2, and type II)
Not color specified (type I, class 3)
Poor dye penetration, mottled, streaky, or cloudy (type I, class 3)
Width not within specified tolerance 1/
Length of panels not within specified tolerance (type II)
Width of pinking bar not within tolerance (type II)
Folding stripes, if present, not in specified locations (type II)
Any piece which does not begin or end with a pinked edge in the middle of the pinking bar (type II)
Not labeled in accordance with the Rules and Regulations Under the Textile Fiber Products Identification Act

1/ Shall not be scored unless it occurs for more than 5 yards per roll.

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TABLE III. Examination criteria

Lot size (yards)	Sample size (rolls)	Maximum number of defects acceptable in sample 2/
Up to 1200 1/	3	0
1201 up to and including 3200	5	0
3201 up to and including 10,000	8	0
10,001 up to and including 35,000	13	1
35,001 up to and including 150,000	20	2
150,001 and over	32	3

1/ If a lot contains fewer than three rolls, each roll in the lot shall be examined.

2/ This column is applicable to 4.2.2.2 and 4.2.2.3.1.

4.2.2.3 Length examination.

4.2.2.3.1 Individual rolls. Each roll selected for yard-by-yard examination in accordance with table III shall be examined for length. Any length found to be less than the minimum specified or more than 2 yards less than the length marked on the ticket shall be considered a defect. If the total number of defects in the sample rolls exceeds the maximum number specified in table III, the lot shall be rejected.

4.2.2.3.2 Total yardage in sample. The lot shall be unacceptable if the total of the actual length of rolls in the sample is less than the total of the lengths marked on the tickets.

4.2.3 End item testing. The methods of testing specified in FED-STD-191, wherever applicable, and as listed in table IV, shall be followed. Except as otherwise indicated, the physical and chemical values specified in section 3 apply to the average of the determinations made on a sample unit for test purposes as specified in the applicable test methods. The sample unit shall be 3 continuous yards full width of the finished cloth and 1/4 yard full width of type I, class 3 cloth prior to dyeing. The lot shall be unacceptable if one or more sample units fail to meet any test requirement specified. All test reports shall contain the individual values utilized in expressing the final results. The sample size (number of sample units) shall be in accordance with the following:

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Lot size (yards)	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

TABLE IV. End item tests

Characteristic	Requirement paragraph	Test method
Material, cotton	3.2.1	1200 1/
Discoloration	3.3.1	5660 2/
Presence of labile sulfur	3.3.2	2020
Colorfastness to		
Laundering (after 3 cycles)	3.3.4	5610 3/
Perspiration	3.3.4	5680
Light	3.3.4	5660
Crocking	3.3.4	5651
Weight	3.4	5041 4/
Yarns per inch	3.4	5050 5/
Breaking strength	3.4	5100 6/
Weave	3.4.1	Visual 7/
Shrinkage in laundering	3.5	5550 8/
Absorbency	3.6	5502
Nonfibrous materials	3.7.1	2611
pH	3.8	2811

1/ Unless otherwise specified, a certificate of compliance is required and will be acceptable for the requirements specified.

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2/ Except that time of exposure shall be for 20 standard fading hours.

3/ The specimens must be dried after each of the three laundering cycles. On the color transfer cloth evaluation, only the stain on the cotton fibers of the color transfer cloth shall be evaluated.

4/ For type II - determination to be made on the two layers of body cloth and should not include folding stripes if present.

5/ For type II - determination to be made on single layer of panel, specimens shall be sampled from both layers and must meet the requirement. The specimens should not include folding stripes if present.

6/ For type II - determination to be made of double layer body cloth (not single ply). All tests must be clear of, and must not include the pinking bars or folding stripes, if present.

7/ One determination shall be made from each sample unit and the results reported as "pass" or "fail".

8/ The specimens of type II cloth shall be a complete diaper, double layer, and shall be cut the width of the cloth through the middle of the respective pinking bars, leaving a pinked edge. The samples shall be dried prior to pressing.

4.2.4 Packaging inspection. An examination shall be made in accordance with the provisions of PPP-P-1134 to determine that packaging, packing, and marking comply with the section 5 requirements.

5. PREPARATION FOR DELIVERY

5.1 Put-up and packaging. Put-up and packaging shall be level A or C as specified (see 6.2).

5.1.1 Levels A and C. The cloth shall be put-up and packaged in accordance with the applicable requirements of PPP-P-1134.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Levels A, B and C. Cloth of one type and class only shall be packed in accordance with the applicable requirements of PPP-P-1134.

5.3 Marking. In addition to any special marking required by the contract or order, shipments shall be marked in accordance with the applicable requirements of PPP-P-1134.

6. NOTES

6.1 Intended use. The cloths are intended for the following:

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(a) Type I, class 1 cloth and type II cloth are intended for use in the fabrication of infant diapers.
(b) Type I, classes 2 and 3 cloths are intended for use in the fabrication of coveralls, cooling, toxicological agents protective and rocket fuel handlers. Class 2 is used by the Air Force and class 3 is used by the Army.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type and class of cloth required (see 1.2.1).
(c) Color of cloth required (see 3.3.2).
(d) Width required (see 3.4).
(e) Minimum length, if other than specified (see 3.9).
(f) Selection of applicable levels of put-up, packaging and packing (see 5.1 and 5.2).

6.3 Standard sample. For access to standard sample, address the procuring activity issuing the invitation for bids.

6.4 Dyestuff formulation.

6.4.1 Olive Green 107. A suggested but not mandatory dyestuff formulation for Olive Green 107 (OG 107) is as follows:

New color index

Vat Black 25, CI 69525
Vat Green 3, CI 69500
Vat Green 8, CI 71050

Shaded with either or both of the following:

Vat Brown 3, CT 69015

Vat Yellow of suitable fastness

The redness of shade should result from the predominant use of the main colors specified and not from the shading colors.

6.5 GSA procurement. For GSA procurement, rolls or bolts shall be 50 yards in length with a tolerance of within plus or minus 2-1/2 yards.

6.6 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification.

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MILITARY INTERESTS:

Custodians

Army - GL
Navy - NU
Air Force - 99

Review Activities

Army - MD
Navy - MS

User Activity

DLA - DM

CIVIL AGENCY COORDINATING ACTIVITIES:

COMMERCE - NBS
GSA - FSS
VA - OSS

PREPARING ACTIVITY:

Army - GL

Project No. 8305-0258

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